3D Brain Tumor Segmentation

AURA — A Hybrid CNN–Transformer Architecture

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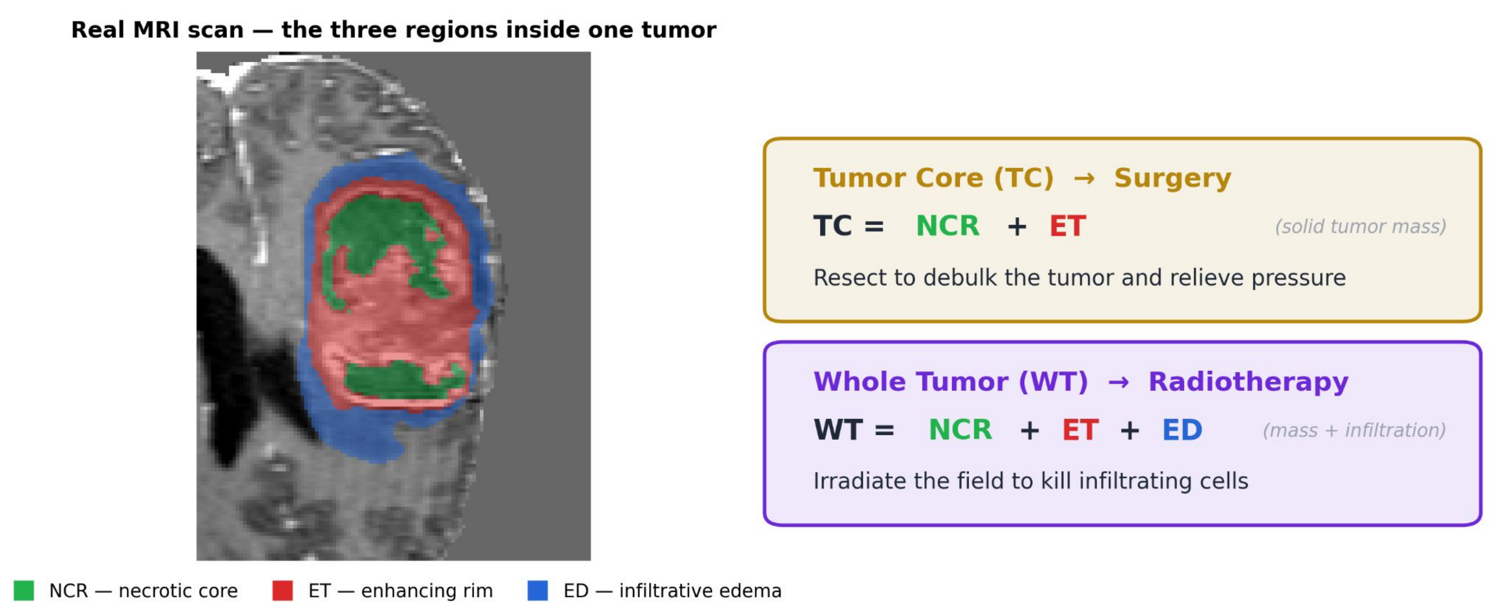
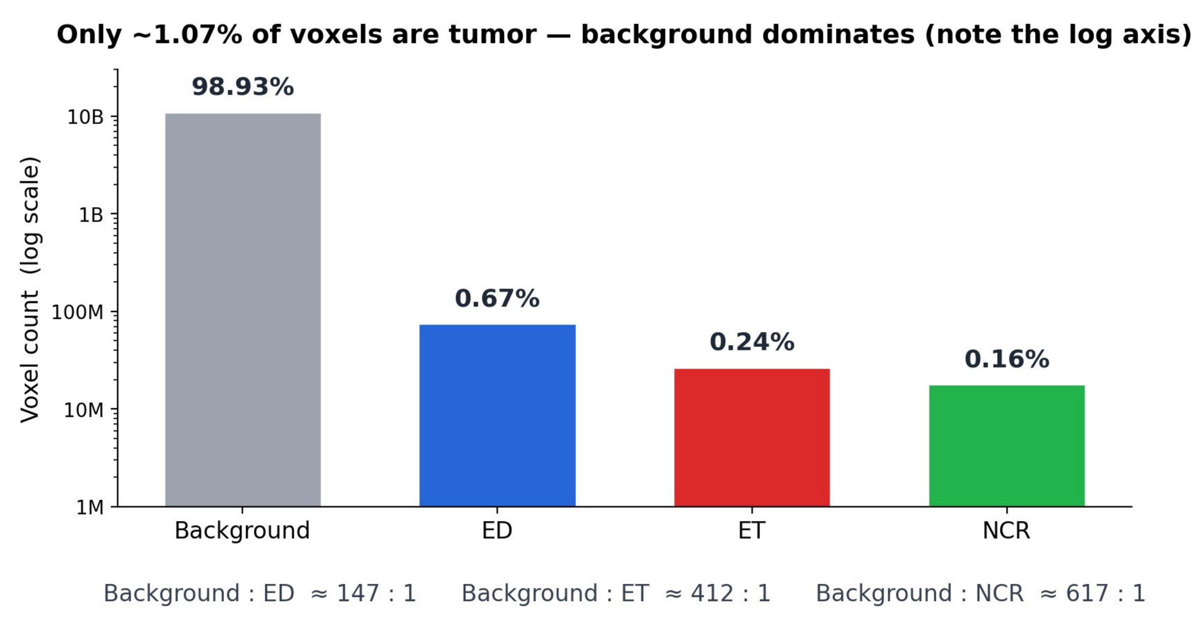
1. Clinical Problem

- Brain tumors have a 5-year survival rate of only ~30%
- Doctors rely on 3 nested regions: WT (radiotherapy) · TC (surgery) · ET (aggressiveness)
- Manual MRI labeling across 4 scans takes up to 4 hours
- No standard tool provides 3D visualization with instant volume metrics
- A useful model must also flag when it is uncertain



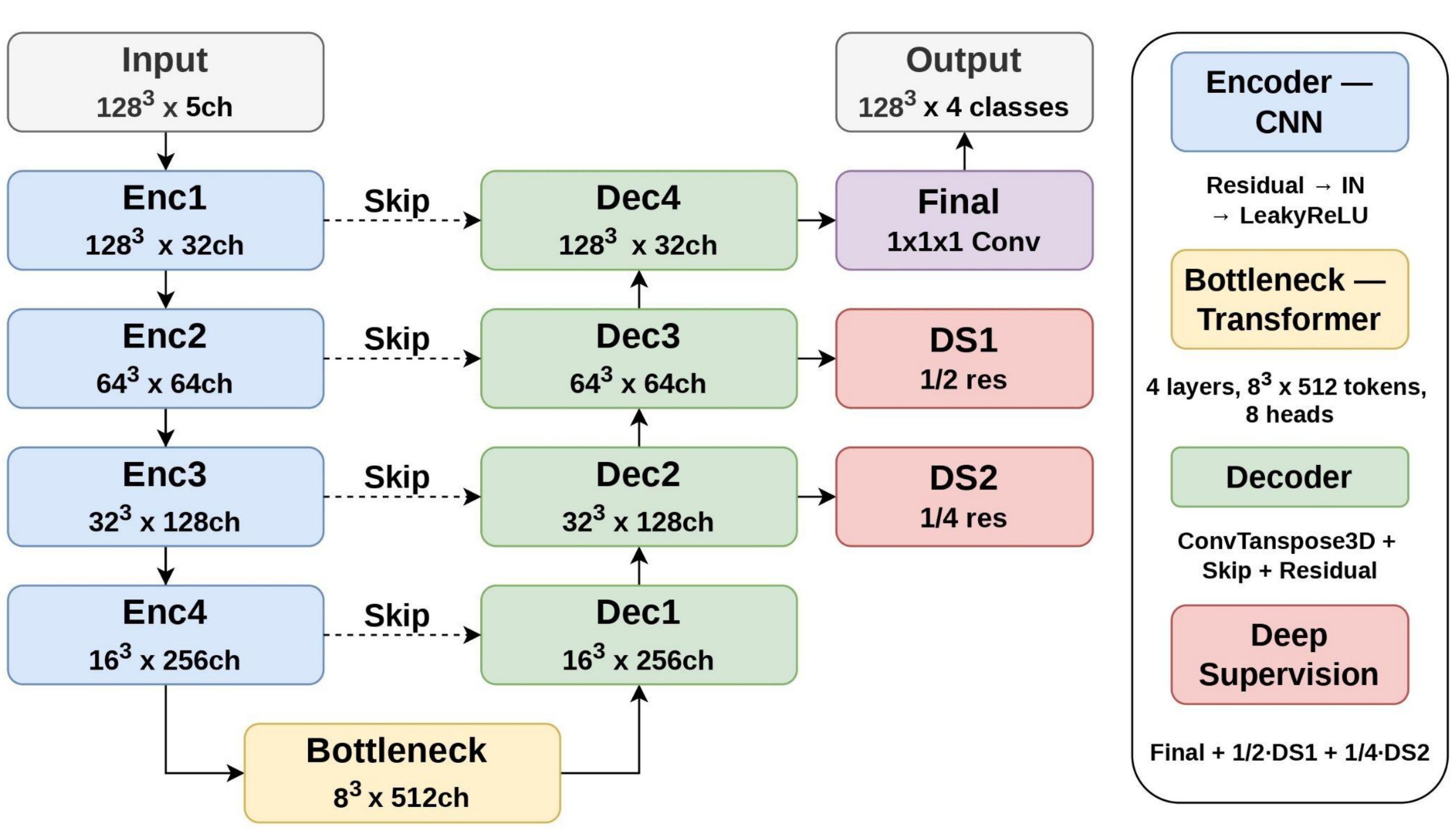
2. Dataset

- BraTS 2021 — 1251 cases, 1000 train / 251 val
- 4 modalities per case: T1 · T1CE · T2 · FLAIR; no single modality shows the full tumor
- Input: 5-channel 128^3 patches (4 modalities + foreground mask)
- Background-to-ET ratio ~412:1; region-wise Dice + Focal loss and 50% tumor-centered patch sampling address the imbalance



3. AURA Architecture

- CNN + Transformer hybrid: ResU-Net encoder–decoder with windowed Transformer at the 8^3 bottleneck
- CNN captures local texture; Transformer adds global 3D context at low cost
- 4-class softmax head avoids the region-overlap ambiguity of sigmoid outputs
- Deep supervision propagates gradients through intermediate decoder stages
- MC-Dropout decoder enables uncertainty estimation at inference with no extra parameters



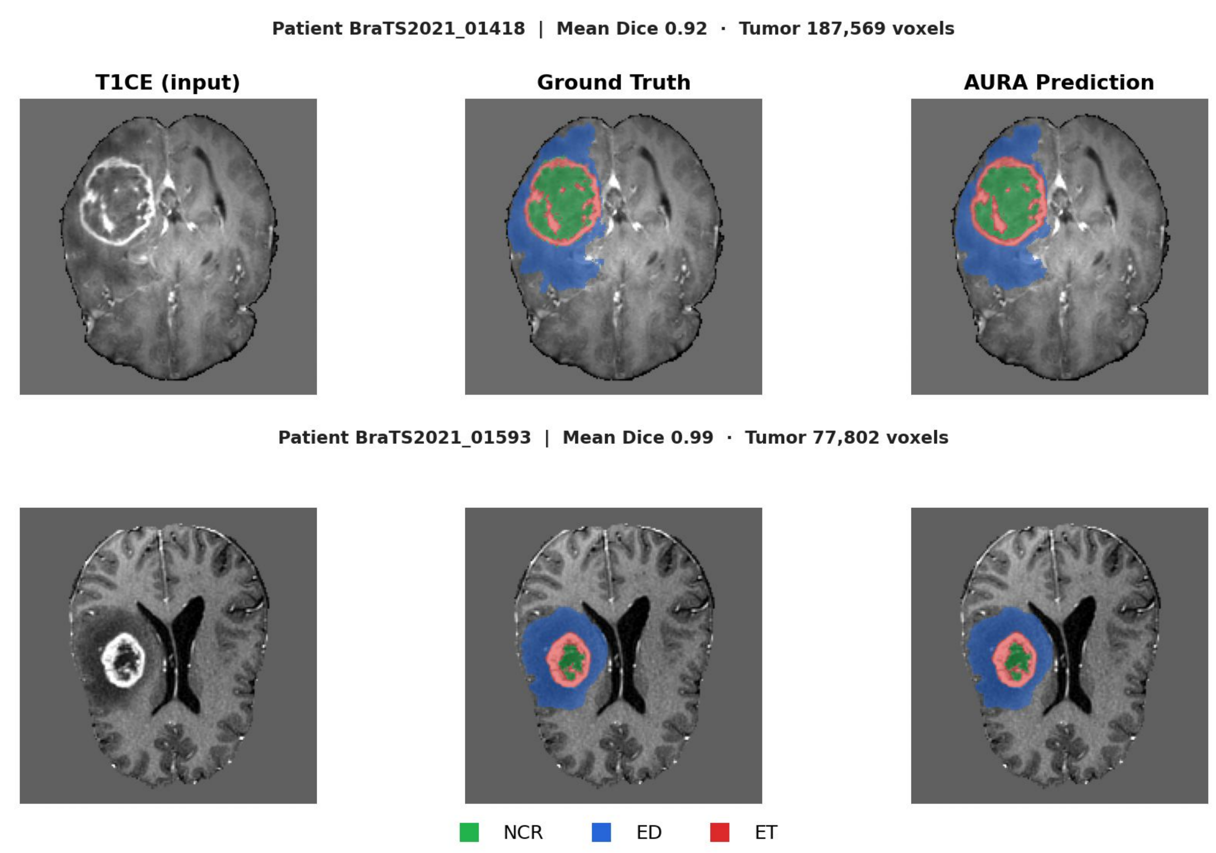
4. Results

Model	Baseline	Complex	AURA
Mean Dice	0.825	0.833	0.839
Dice ET	0.765	0.789	0.785
Dice TC	0.788	0.781	0.804
Dice WT	0.923	0.929	0.928
HD95 ↓	6.75	5.95	5.96
ECE ↓	0.106	0.110	0.100
AURC ↓	0.179	0.191	0.175

- AURA achieves the highest Mean Dice with only 2 modules vs. 8 in Complex
- Wins TC by +0.023 over Complex — the most clinically critical region
- Best calibration (ECE 0.100) and uncertainty quality (AURC 0.175) across all models

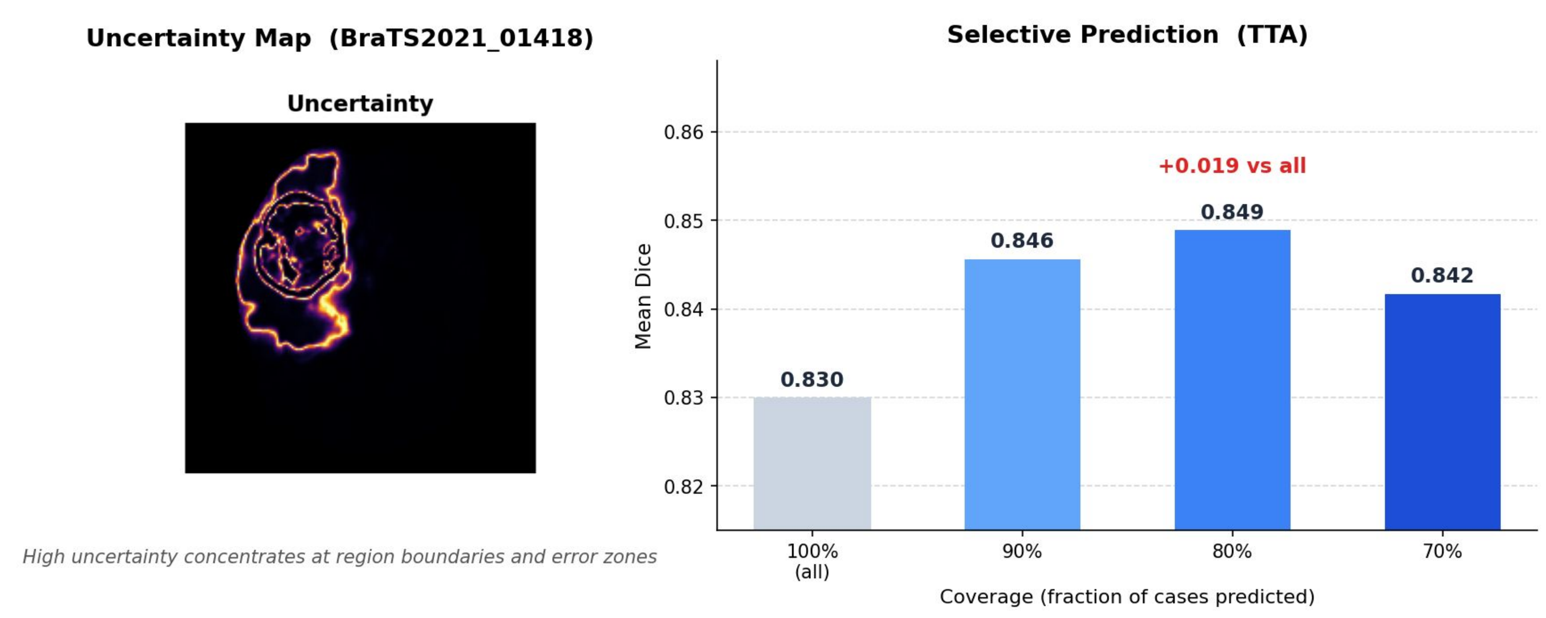
5. Qualitative Results

- AURA correctly recovers all three tumor regions including fine boundaries and the small enhancing core
- Fails when ET is very small or NCR is visually ambiguous — tends to misclassify NCR as edema
- Uncertainty is highest exactly where the model fails, confirming it as a reliable error signal



6. Trustworthiness

- AURA outputs per-voxel uncertainty maps; high uncertainty concentrates at region boundaries and error-prone cases
- Skipping the least-confident 20% raises Mean Dice from 0.830 → 0.849 (+0.019)
- ECE 0.100 — predicted confidence closely matches observed accuracy on tumor voxels



7. Complexity

Metric	AURA
Parameters	45 M
FLOPs	612 GFLOPs
GPU Memory	1.9 GB
Model Size	172 MB
Inference Time	242 ms
Throughput	4.1 cases/s

8. Deployment

- Deployed as a FastAPI + Niivue browser workstation
- Upload 4 NIfTI files → 3D overlay · tumor volumes · per-region confidence · anatomical location
- Exports a one-paragraph clinical summary as a ZIP report

9. Conclusion

AURA delivers accurate, calibrated, and uncertainty-aware brain tumor segmentation from a simple CNN + Transformer design — outperforming a far more complex model and deployable on a single consumer GPU.